MAHATMA GANDHI UNIVERSITY, KOTTAYAM
Industrial Chemistry Minor
SEMESTER III
Model Question Paper
Course Code: MG3DSCICH200
Course Title: Functional Operations in Chemical Industry

Time: 1.5 Hours **Maximum Marks**: 50

Section A

(Answer all questions. Each carries 1 mark.)

1. The flow in which fluid layers slide past each other without mixing is called *[CO1, U]*
A) Turbulent flow B) Laminar flow C) Compressible flow D) Viscous flow

2. The equipment primarily used for separating liquids based on boiling points is: *[CO1, K]*
A) Evaporator B) Extractor C) Crystalliser D) Distillation column
3. Reynolds number is an indicator of: *[CO1, U]*
A) Viscosity B) Flow regime C) Temperature D) Surface area

4. Which of the following is an example of an evaporator used in industry? *[CO1, U]*
A) Spray dryer B) Rotary evaporator C) Falling film evaporator D) Drum dryer

5. The Czochralski method is associated with: *[CO2, U]*
A) Solvent extraction B) Crystal growth C) Drying D) Distillation

6. The principle of solvent extraction is based on: *[CO3, U]*
A) Boiling point difference B) Solubility difference C) Pressure difference D) Surface tension

7. Which of the following is not a mixing equipment? *[CO3, U]*
A) Paddle mixer B) Trough mixer C) Flash evaporator D) Tumbler mixer

8. Steam distillation is mainly used for separating: *[CO1, U]*
A) Volatile immiscible liquids B) Inorganic salts C) Polymers D) Gases

9. Why is an MSMPR crystalliser preferred over a batch crystalliser in large-scale continuous production? *[CO2, An]*
A) It occupies less space and costs less.
B) It can handle multiple solutes simultaneously.
C) It maintains steady-state operation, allowing uniform crystal size and continuous output.
D) It avoids the need for temperature control.

10. The Verneuil method is a crystal growth technique used to produce: *[CO2, A]*
A) Sugar crystals B) Benzoic acid crystals C) Synthetic gemstones D) Proteins

(10 × 1 = 10 marks)

Section B – Short Answer Questions

Answer any four questions. Each carries 3 marks.

11. Define Bernoulli's principle. How is it applicable in fluid flow operations? *[CO1, U]*
12. Differentiate between laminar and turbulent flow with one example each. *[CO1, U]*
13. Explain the principle behind fractional distillation. *[CO1, U]*
14. Compare the Bridgman and Czochralski methods of crystal growth in terms of heat transfer mechanism and control over crystal quality. [CO2, An]
15. What are the basic steps involved in crystallisation? *[CO2, U]*
16. Define supercritical fluid extraction and mention one industrial application. *[CO3, U]*

(4 × 3 = 12 marks)

Section C – Short Essay Questions

Answer any four questions. Each carries 7 marks.

17. Describe the working of falling film and climbing film evaporators with diagrams. *[CO1, U]*
18. Explain the different types of distillation methods used in chemical industries. *[CO1, U]*
19. Discuss in detail the principle and techniques of crystal growth using Bridgman and Czochralski methods. *[CO2, An]*
20. Describe the construction and working of MSMPR and draft tube crystallisers. *[CO2, U]*
21. Explain the process of liquid-liquid extraction and solid-liquid extraction with suitable examples. *[CO3, U]*
22. Compare the mixing mechanisms for solid-solid and liquid-liquid systems. Mention the mixers used in each. [*CO3, U]*

(4 × 7 = 28 marks)